

Analyzing LinkedIn Job Postings

IT 462 – Big Data Systems

Dr. Afshan Jafri

Prepared By

Ghadeer Alnuwaysir | 444200420

Jana Alruzuq | 444201157

Rana Albridi | 444201094

Ghena Almogayad | 444203140



Introduction

The rapid growth of online job platforms has transformed how labor market data is generated and analyzed. Platforms like LinkedIn provide large-scale, real-time data on job postings, including salaries, roles, and locations. This creates an opportunity to apply big data analytics to understand market trends and extract meaningful insights.



Problem and Motivation



Despite the availability of job posting data, salary information remains inconsistent, incomplete, and difficult to interpret. This makes it challenging to understand compensation trends or make informed decisions. Therefore, this project aims to analyze the key factors affecting salary levels and build a predictive model to estimate salaries based on job attributes, supporting job seekers, recruiters, and policymakers.

Data

Dataset

- LinkedIn Job Postings (2023–2024)
- Source: Kaggle
 - Size: 124,000+ records (~500MB)
 - Type: Structured job market data

Main Features

- Job title
- Salary (min, max, median)
- Location
- Work type

Why?

- Large-scale dataset
- Real-world data
- Requires Spark for processing

LinkedIn Job Postings (2023 - 2024)

▲

450

<> Code

Data Card

Code (53)

Discussion (35)

Suggestions (0)

Detail

Compact

Column

10 of 31 columns ▼

🔗 job_id	🔗 company_...	🔗 title	🔗 description	# max_salary	🔗 pay_period	
921716	Corcoran Sawyer Smith	Marketing Coordinator	Job descriptionA leading real estate firm in New Jersey is	20.0	HOURLY	

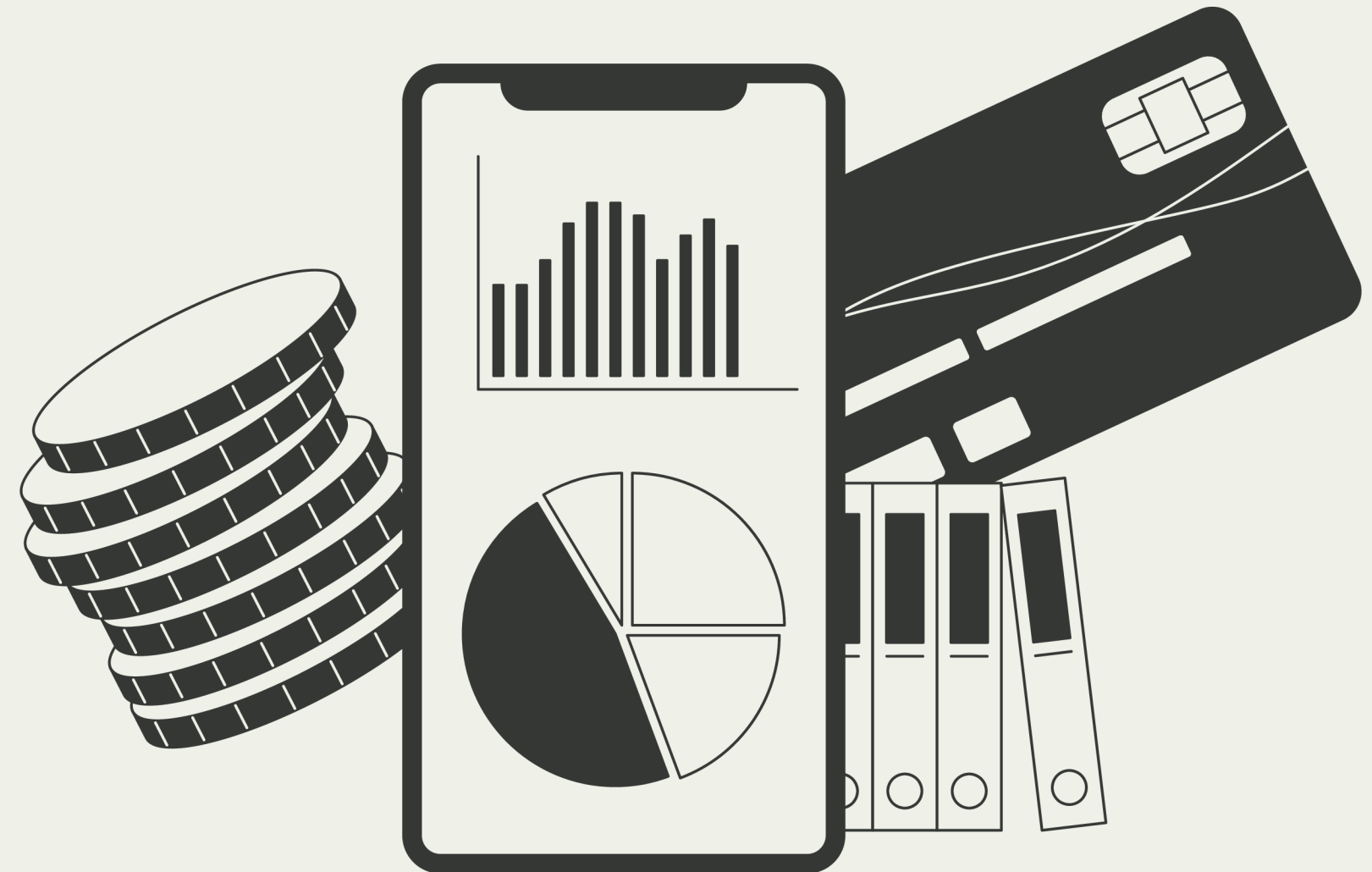
Challenges

Main Challenges

- ~70% missing salary data
- Extreme salary outliers
- Mixed pay formats (hourly, monthly, yearly)
- Duplicate job postings

Impact

- Required filtering & standardization
- Reduced dataset size significantly



Preprocessing

1

Cleaning

- Removed missing critical values
- Handled null engagement data

2

Reduction

- Removed unnecessary columns
- Filtered rows with valid salaries

3

Transformation

- Log transformations (salary, views, applies)
- Extracted time features
- Created location & title features

Step	Before	After
Raw dataset	123,849 rows	122,130 clean rows
Feature selection	33 columns	20 columns
Salary-ready ML data	35,562 rows	35,118 rows

RDD Exploration

KEY RDD OPERATIONS

Salary Filtering: used `filter` and `cache` to keep only full-time postings with valid salary values

Salary Tiers: used `map` and `reduceByKey` to group salaries into LOW, MID, and HIGH

Job Title Keywords: used `flatMap`, `reduceByKey`, and `sortBy` to find the most common job-title words

✦ RDDs clarified salary patterns and key job-market trends.

Code Snippet

```
val rdd_salary_bucket =  
rdd_fulltime_salary.map { row =>
```

```
    val salary = Option(row.getAs[Any]  
("normalized_salary")).map(_.toString.t  
oDouble).getOrElse(0.0)
```

```
    val bucket = if (salary < p33) "LOW"  
else if (salary < p67) "MID" else "HIGH"  
    (bucket, 1)
```

```
}.reduceByKey(_ + _)
```

Result:

- LOW: 9,487 postings
- MID: 9,772 postings
- HIGH: 9,490 postings

SQL Analytics Highlights

1

Salary by work type:

full-time yearly jobs showed the most stable salary patterns

2

Top states by salary:

KY, SC, and TX had the highest average salaries

3

Engagement by work type:

contract roles received the highest candidate interaction

These SQL results helped justify the ML features:

state, **work type**, **experience level**, **job title**, **views**, and **applications** were included because they showed clear salary-related patterns during analysis

EXAMPLE RESULT

<i>rank</i>	<i>location</i>	<i>avg salary</i>	<i>postings</i>
1	KY	\$750,234	1,172
2	SC	\$665,633	1,533
3	TX	\$656,746	10,197
4	OK	\$599,642	790
5	MN	\$361,021	1,816

- **High job volume does not always mean high salary.**
- **Location has a clear effect on compensation.**
- **Smaller markets can still show higher average salaries.**

ML Approach

Problem

- **Type:** Regression
- **Target:** `log_salary`

Training Setup

- Split: 70% train / 30% test
- Seed: 42

Models Used

- **Linear Regression:** simple baseline for salary prediction.
- **Random Forest:** handles mixed features and non-linear salary patterns.
- **Gradient Boosted Trees:** captures stronger feature interactions and usually performs well on structured data.

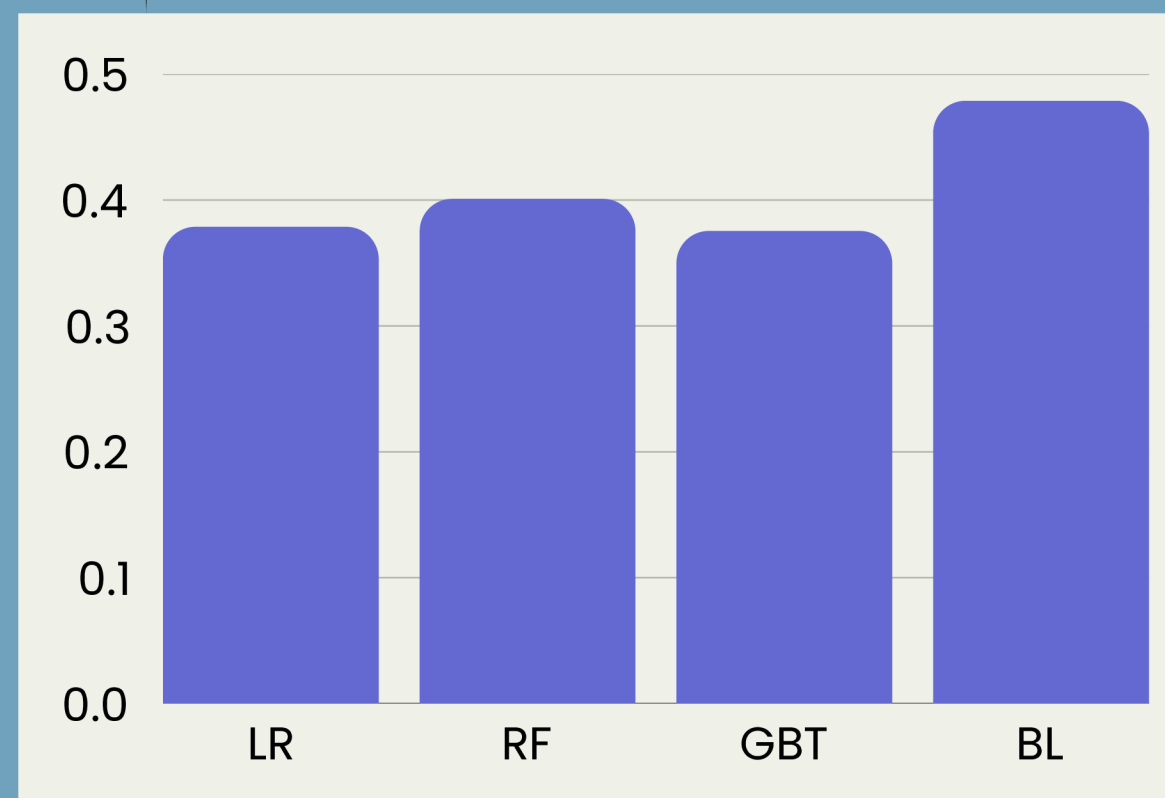
Key Features

- ✦ **Job title:** title_len, title_words, TF-IDF terms
- ✦ **Location:** state
- ✦ **Experience level + work type**
- ✦ **Engagement:** log_views, log_applies

Results & Model Performance

Model	RMSE	MAE	R ²
Baseline	0.4788	0.3814	0
Linear Regression	0.3787	0.2841	0.3745
Random Forest	0.4009	0.3075	0.2989
GBT	0.3754	0.2744	0.3852

✦ **RMSE by Model**



GBT reduced prediction error by about **22%** compared to the baseline, making it the **best-performing** salary prediction model

Interpretation & Insights

Important Features

- **Job title length matters:** longer and more specific titles often indicate senior or specialized roles with higher salaries.
- **Location is a strong salary driver:** some states show higher average salaries even with fewer postings.
- **Experience level affects compensation:** senior roles generally align with higher salary predictions.

Interpretation & Insights

Model Performance

- **Model performs best on middle-range salaries:** predictions are more reliable for common salary ranges.
- **Model struggles with very low salaries:** it tends to overestimate them due to limited low-salary examples.

Implication: job seekers can compare salaries by **role** and **location**; recruiters can benchmark offers more realistically.

Limitations & Future Work

Limitations

- Many postings had missing or inconsistent salary values.
- Salaries were self-reported, which adds noise.
- Limited feature depth: company size, industry, and skills were not fully used.

Future Work

- Add company-level features such as industry and company size.
- Use better text features from job titles/descriptions, such as Word2Vec or BERT.
- Apply hyperparameter tuning to improve model performance.



The model is useful for salary estimation, but results should not be treated as exact salary predictions.

Final Conclusion

- **Cleaned and transformed a large LinkedIn job postings dataset.**
- **Used RDD and SQL analysis to explore salary and job-market patterns.**
- **Built regression models to predict salary levels.**

Salary prediction is possible, but it depends heavily on job title, location, experience level, work type, and data quality.



**Do You Have Any
Questions?**

Thank You

